Exp 4: Data Modelling and Form Design for an Insurance Claim Application in Zoho Creator (Advanced Fields)

1. Aim

To extend the existing “Insurance Claim Request Form” application in Zoho Creator by incorporating all seventeen Advanced Fields available in the Form Builder, so that the form moves beyond basic data capture and demonstrates automation, computed values, file/media handling, and structured lookups within a real insurance-servicing use case — from policy verification and witness capture through settlement computation and the issue of a scannable claim tracking reference.

2. Procedure

2.1 Problem Statement & Application Scope

The basic Claim Request Form (built earlier using the thirteen Basic Fields) captures a claimant’s identity, contact details, and incident description, but it cannot handle several real-world requirements of an insurance branch: verifying policy details against the master policy database, recording multiple witness statements for a single incident, computing the net payable amount after depreciation and deductibles, issuing a unique scannable claim tracking reference, or accepting supporting media such as damage photographs, a repair estimate document, or a signed declaration.

The application designed here, **Insurance Claim Request & Processing Management (Advanced Edition)**, extends the *Claim_Request_Form* so that, in addition to the existing basic fields, it uses every Advanced Field type provided by Zoho Creator’s Form Builder. The primary users are Policyholders, Nominees and the Claims Processing Team. The extended report view, *Claim_Request_Report*, adds computed settlement, claim reference and QR code columns, enabling the claims team to verify policies automatically, record witness statements, compute net payable amounts, and issue scannable digital claim tracking references.

2.2 Data Model & ER Diagram

The diagram below shows the internal schema built for this application: the primary form entity with its Advanced fields, the Witness_Details SubForm capturing repeating witness rows, the Policies_Master table that the Policy_Details_Lookup field draws from, and the system-generated values derived from the Claim Reference ID.

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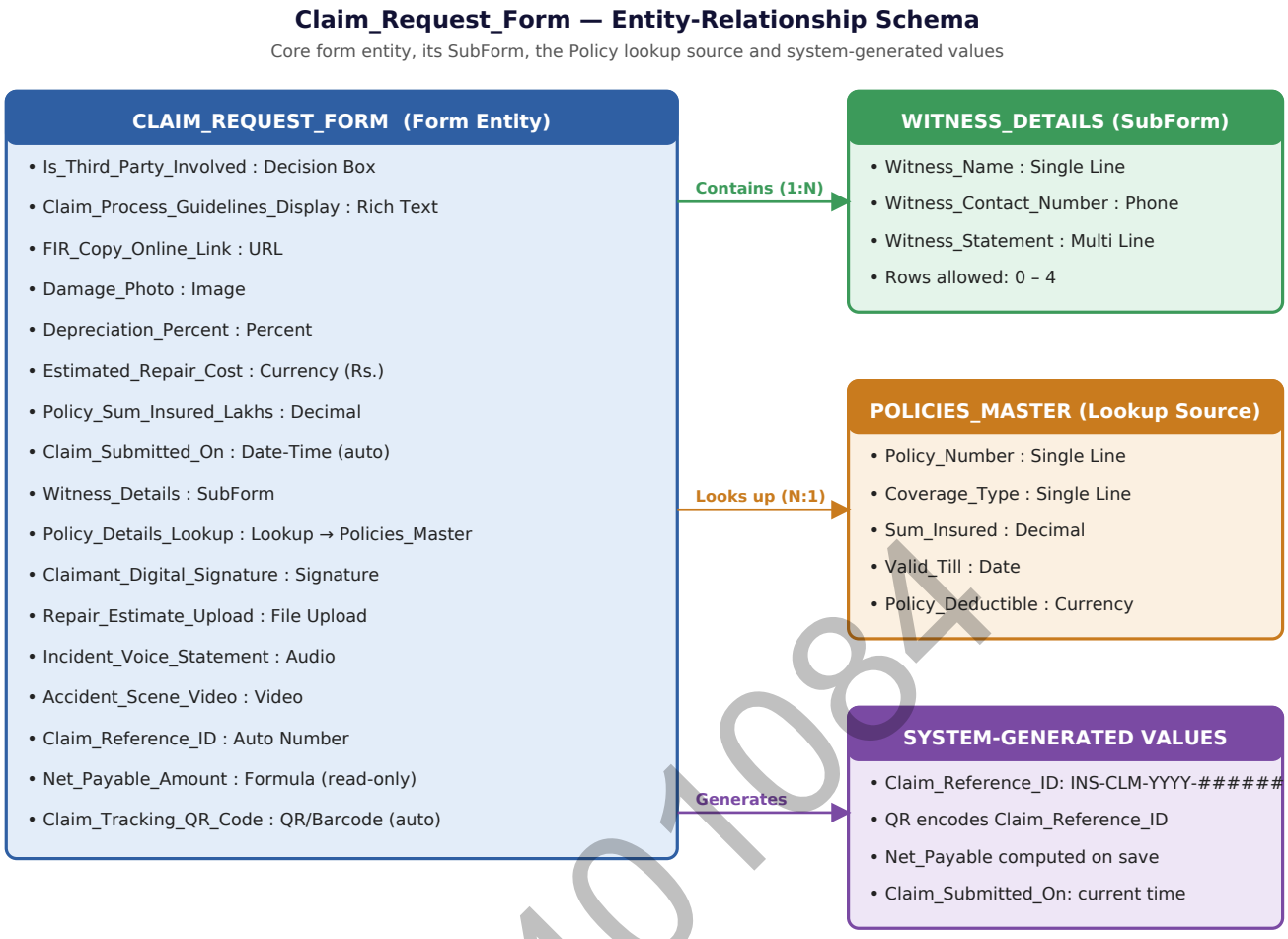


Figure 1: Entity-Relationship schema for Claim_Request_Form

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2.3 Advanced Fields Mapping & Data Dictionary

Table 1 maps every Advanced Field from the Zoho Creator Form Builder palette to its role in the extended Insurance Claim Request Form, along with sample values and behaviour.

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Decision Box	Is_Third_Party_Involved	Captures a quick Yes/No decision on whether a third party was involved in the incident.	Yes / No toggle
Rich Text	Claim_Process_Guidelines_Display	Displays formatted claim process guidelines (bold headings, bullet points, hyperlinks) directly within the form.	“Keep all original bills and estimates. A surveyor will contact you within 48 hours. Do not repair the vehicle before inspection.”
URL	FIR_Copy_Online_Link	Captures a link to an online-hosted copy of the police FIR or incident report, where applicable.	https://efir.tnpolice.gov.in/case/88231
Image	Damage_Photo	Captures a photograph of the damaged property or vehicle for preliminary assessment.	damage_photo_01.jpg
Percent	Depreciation_Percent	Depreciation percentage applicable to the claimed part or asset as per policy terms.	20%

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Currency	Estimated_Repair_Cost	Estimated cost of repair or replacement, formatted with currency symbol.	₹38,000.00
Decimal	Policy_Sum_Insured_Lakhs	Sum insured under the policy in lakhs, used to check the claim against policy limits.	5.50
Date-Time	Claim_Submitted_On	Automatically records the exact date and time the claim request was submitted.	24-Jul-2026 16:20
SubForm	Witness_Details	Repeating rows of witness name, contact number, and statement for the reported incident.	Row 1: Mohan K. – 9876543210 – Saw the collision from the roadside; Row 2: Kavya S. – 9123456780 – Confirmed signal was red
Lookup	Policy_Details_Lookup	Looks up and auto-fills the policyholder's coverage type, sum insured, and validity from a master Policies table.	POL-2024-778812 – Comprehensive – Valid till 31-Mar-2027
Signature	Claimant_Digital_Signature	Captures the claimant's digital signature confirming the declaration of authenticity.	Signed (drawn signature captured)
File Upload	Repair_Estimate_Upload	Allows the claimant to upload a scanned copy of the garage/repair estimate or medical bill.	repair_estimate.pdf
Audio	Incident_Voice_Statement	Allows the claimant to optionally record a short voice statement describing the incident.	voice_statement_01.mp3 (0:35)
Video	Accident_Scene_Video	Allows upload of a short video of the accident scene or damaged property for the surveyor's reference.	accident_scene.mp4 (0:50)
Auto Number	Claim_Reference_ID	Automatically generates a unique, sequential claim reference number for every submission.	INS-CLM-2026-000384
Formula	Net_Payable_Amount	Automatically computes the net settlement amount using the repair cost, depreciation, and policy deductible.	Estimated_Repair_Cost − (Estimated_Repair_Cost × Depreciation_Percent) − Policy_Deductible
QR/Barcode	Claim_Tracking_QR_Code	Automatically generates a scannable QR code encoding the Claim Reference ID for status tracking at branch counters.	QR image encoding INS-CLM-2026-000384

Table 1: Mapping and specification of the seventeen Advanced Fields in Claim_Request_Form

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2.4 Form Layout & Configuration

The sections below were added to the existing Claimant, Contact, and Claim Details sections of the basic Insurance Claim Request Form. The four new sections group the Advanced Fields by the stage of the claim they support.

Section E — Policy Verification & Evidence: Policy Details Lookup (Lookup, auto-fills from master Policies table), Damage Photo (Image), Repair Estimate Upload (File Upload), Policy Sum Insured in Lakhs (Decimal).

Section F — Witness & Media Details: Witness Details (SubForm, repeating rows), FIR Copy Online Link (URL), Accident Scene Video (Video, optional), Incident Voice Statement (Audio, optional).

Section G — Settlement Computation: Is Third Party Involved (Decision Box), Estimated Repair Cost (Currency), Depreciation Percent (Percent), Net Payable Amount (Formula, read-only, auto-computed).

Section H — Declaration & System-Generated Values: Claim Process Guidelines Display (Rich Text, read-only reference block), Claimant Digital Signature (Signature), Claim Submitted On (Date-Time, auto-captured), Claim

Reference ID (Auto Number, system-generated), Claim Tracking QR Code (QR/Barcode, system-generated from Claim Reference ID).

2.5 Validation & Configuration Rules

The following field-level rules were configured in the Field Properties inspector of each Advanced Field:

Field Type	Field	Configuration
Lookup	Policy_Details_Lookup	Connected to the master Policies form; claims staff type-ahead search and select rather than typing free text.
SubForm	Witness_Details	Minimum 0 and maximum 4 rows per claim; each row holds Witness_Name, Witness_Contact_Number and Witness_Statement.
File Upload	Repair_Estimate_Upload	Accepts PDF/JPG/PNG scans of the garage estimate or medical bill, with a maximum file size of 5 MB.
Image	Damage_Photo	Accepts JPG/PNG photographs only, with a maximum file size of 2 MB.
Percent	Depreciation_Percent	Value restricted to the range 0–100.
Currency	Estimated_Repair_Cost	Indian Rupee (₹) symbol with two decimal places; must be greater than zero.
Decimal	Policy_Sum_Insured_Lakhs	Two decimal places (e.g. 5.50); auto-filled from the selected policy.
Signature / Video	Claimant_Digital_Signature, Accident_Scene_Video	Marked mandatory on the form, as shown in the form preview (Figure 3).
Formula	Net_Payable_Amount	Read-only; computed on save as Estimated_Repair_Cost − (Estimated_Repair_Cost × Depreciation_Percent / 100) − Policy_Deductible.
Auto Number / QR	Claim_Reference_ID, Claim_Tracking_QR_Code	Auto Number uses prefix INS-CLM- + year + 6-digit sequence; the QR code is generated from the Claim_Reference_ID value.

Table 2: Validation and configuration rules for the Advanced Fields

2.6 Workflow Scripting Configuration

A Deluge validation script is configured under Workflows > Form Workflows > Validations on form submission to enforce business-logic checks before a claim is accepted: the depreciation percentage must lie between 0 and 100, the estimated repair cost must be positive and must not exceed the policy sum insured, and no more than four witness rows may be entered.

```
if(input.Depreciation_Percent < 0 || input.Depreciation_Percent > 100)
{
    alert "Depreciation must be between 0 and 100 percent.";
    cancel submit;
}
if(input.Estimated_Repair_Cost <= 0)
{
    alert "Estimated Repair Cost must be greater than zero.";
    cancel submit;
}
if(input.Estimated_Repair_Cost > (input.Policy_Sum_Insured_Lakhs * 100000))
{
    alert "Estimated Repair Cost exceeds the policy sum insured.";
    cancel submit;
}
if(input.Witness_Details.count() > 4)
{
    alert "A maximum of 4 witnesses can be recorded per claim.";
    cancel submit;
}
```

Figure 2: Deluge validation script for Claim_Request_Form (On Validate event)

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3. Output

The following figures show the live preview of the extended Claim_Request_Form in the Zoho Creator Form Builder, covering all of the Advanced Fields — including the Witness_Details SubForm, the Percent, Currency and Decimal fields, the Signature pad, and the File, Audio and Video upload fields.

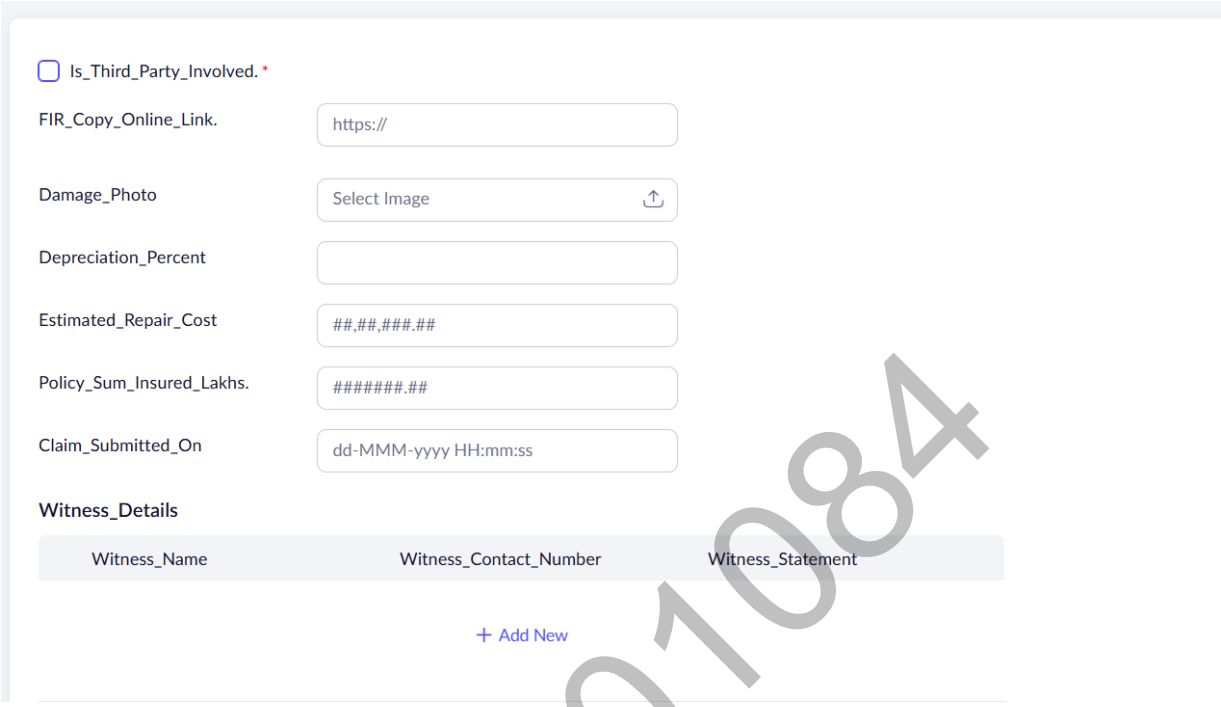


Figure 3: Form Builder preview showing the Decision Box, URL, Image, Percent, Currency, Decimal and Date-Time fields along with the Witness_Details SubForm (Name, Contact Number, Statement)

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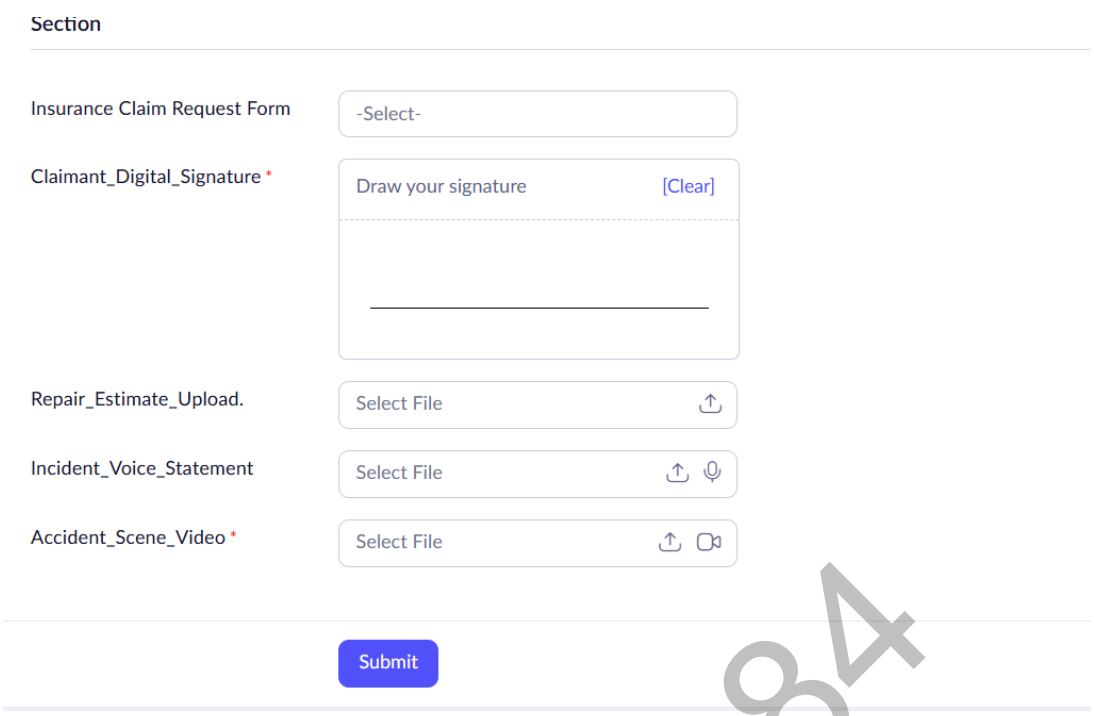


Figure 4: Form Builder preview showing the Claimant_Digital_Signature pad, Repair_Estimate_Upload, Incident_Voice_Statement (with microphone recorder) and Accident_Scene_Video (with video recorder) fields, followed by the Submit button

4. Result

The “Insurance Claim Request & Processing Management (Advanced Edition)” application was successfully extended in Zoho Creator using all seventeen Advanced Fields — Decision Box, Rich Text, URL, Image, Percent, Currency, Decimal, Date-Time, SubForm, Lookup, Signature, File Upload, Audio, Video, Auto Number, Formula and QR/Barcode — with field-level validations, a four-section extended layout (Sections E–H), a Witness_Details SubForm limited to four rows, a Policies lookup, an auto-computed Net Payable Amount, and automated Claim Reference ID / QR code generation for tracking — meeting all requirements of the advanced field use case.

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